

🔧 The Core Technical Stack of Introvert

To understand **Introvert** from the ground up, think of it as a complete ecosystem that eliminates corporate middlemen. Instead of relying on massive servers to verify your identity, store your chat history, and track your data, Introvert uses a localized architecture powered by native Rust code, local database encryption, and decentralized routing.

The diagram below visualizes how the Flutter UI links to our underlying native tools:

1. 🦀 The Rust Core Engine & Dart FFI

- **What it is:** **Rust** is a high-performance, memory-safe, and secure programming language that drives our background backend tasks. **Dart FFI** acts as a zero-overhead native software bridge that links the Flutter user interface directly to the underlying Rust codebase.
 - **Why we use it:** To guarantee that high-overhead cryptographic processes, localized database queries, and peer-to-peer network streaming run at native hardware speed without lag or memory leaks. Rust handles the heavy lifting, while Flutter constructs the visual interface.
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2. 🔑 Single-Seed Identity (Ed25519)

- **What it is:** A single 12-word cryptographic seed phrase that constructs the master operational keys for the device.
 - **Why we use it:** Instead of requiring email addresses, names, passwords, or phone numbers, your entire node identity is derived using pure mathematics. This single seed acts as your P2P chat identity, creates the key to unlock your local disk storage, and serves as your on-chain Solana wallet.
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3. 🛡️ SQLCipher (Encrypted Local Storage)

- **What it is:** A highly secure, encrypted version of a standard embedded database.
 - **Why we use it:** Since there are no central servers to hold your data, your conversation histories must stay directly on your local disk. SQLCipher ensures that all of your data is securely encrypted; even if someone gets physical access to your device, they cannot read your chat files without your master seed key.
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4. 🌐 libp2p & Kademlia DHT (The Mesh Network)

- **What it is:** **libp2p** is a modular framework for decentralized peer-to-peer networking, and the **Kademlia DHT** is a Distributed Hash Table that works like a decentralized network phone book.
 - **Why we use it:** Instead of pushing your messages through a corporate server (such as WhatsApp or Telegram), your node connects directly to other trusted peers. The Kademlia DHT helps your application resolve exactly where your friends are routed on the internet without exposing their real IP addresses to the network.
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5. 🔒 The Double Ratchet (End-to-End Encryption)

- **What it is:** A dynamic end-to-end encryption (E2EE) protocol that derives new encryption keys for *every single message* that is sent or received.
- **Why we use it:** It guarantees total message privacy. Even if someone manages to compromise a single temporary message key, they still cannot decode your past chats (Forward Secrecy) or read your future messages (Break-in Recovery).

6. 🟡 The Introvert Token (INTR)

- **What it is:** A custom SPL (Solana Program Library) token minted on the Solana Mainnet that acts as our localized reward economy.
- **Why we use it:** To maintain a highly persistent P2P network, we rely on independent, high-uptime "Anchor Nodes" (run by desktop users) that act as digital store-and-forward mailboxes for offline peers. The **INTR** token functions as an automated payout layer—rewarding those Anchor nodes directly for their server uptime and message caching services.